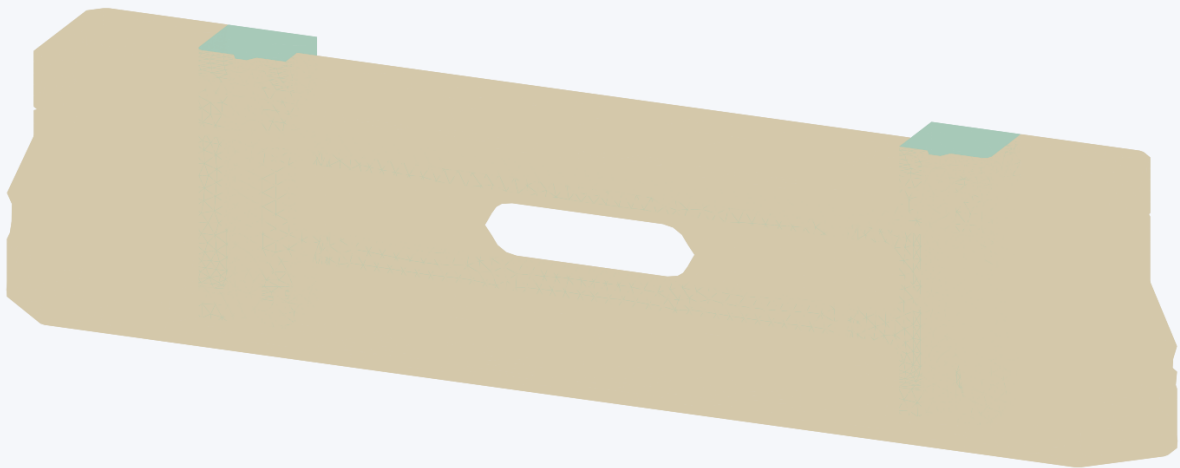


Selected-run evidence status: completed_unassessed

Geometry

Undeformed geometry

Exterior shape from the selected mesh



Mesh-derived exterior • source geometry unchanged

Undeformed geometry from selected mesh. Exterior shape derived from the selected run's mesh; this is not a CAD render.

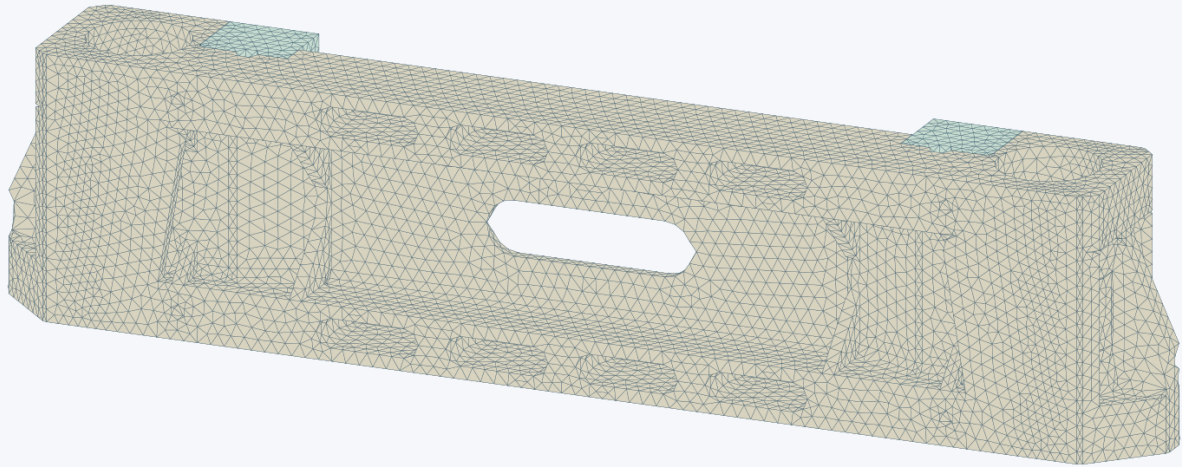
Mesh

Mesh overview and local detail

Mesh nodes	19208	
Tetrahedral elements	67240	
Meshed parts	2	
Nominal element size	4.0	mm
Minimum mean-ratio quality	0.0774	0-1

Mesh overview

Undeformed exterior mesh

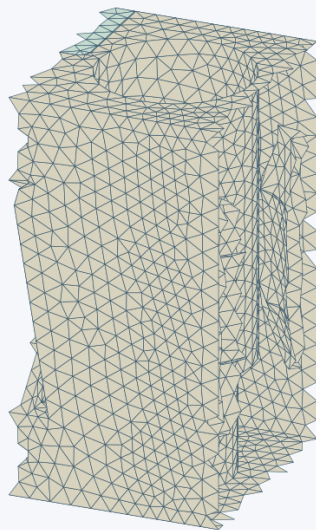


Exterior mesh faces • undeformed geometry

Mesh overview. Undeformed exterior topology of the selected case, verified against its recorded solved-mesh hashes.

Pad-root refinement detail

Pad-root region



Exterior mesh faces • undeformed geometry

Pad-root refinement detail. Positive pad-root region, $X = 112\text{--}150\text{ mm}$. Mesh metadata records a 0.65 target factor (2.6 mm local target).

Mean-ratio quality describes element shape and does not establish stress or time-step convergence.

Setup

Recorded selected-case model

Initial impact velocity	50	m/s
Requested simulation duration	8	ms
Foam depth	203.2	mm
Foam width	41.275	mm
Foam height	152.4	mm
Foam element size	10	mm
Initial contact gap	0.1	mm
Contact friction coefficient	0.1	dimensionless
Foam curve scale	1	dimensionless
Time-step scale	0.6	dimensionless
Requested state-output interval	0.01	ms
Startup mass-scaling DT2MS	-1.5e-07	s
Motor mass	0.1616	kg
Moving mass	0.6503408	kg
Pusher mass	0.4887408	kg
Initial kinetic energy	812.926	J

Recorded material, contact and boundary assumptions

- Body and bolted cable cap flexible; four local perfectly clamped bolt collars. No bolt preload, slip or failure.
- Motor loaded upper-bound inertia distributed over actual bore nodes; no motor housing stiffness or exact inertia tensor.
- PA12-CF10 elastic isotropic proxy $E=1806.6\text{MPa}$ $\nu=.35$ density 1060; print +Z motor axes. No strength/failure claims.
- No fracture, erosion, motor thrust, carrier, released endcap or invented 35g hardware allowance.
- Rigid 8mm aluminum spreaders initially stationary; fixed x,y and rotations, free z without travel limit.
- Front and rear pusher bore-quadrant averages follow fixed zero-clearance coaxial guide axes through eight disjoint LINEAR_GLOBAL X/Y equations; every Z DOF is free. Cable compliance and clearance are

idealized.

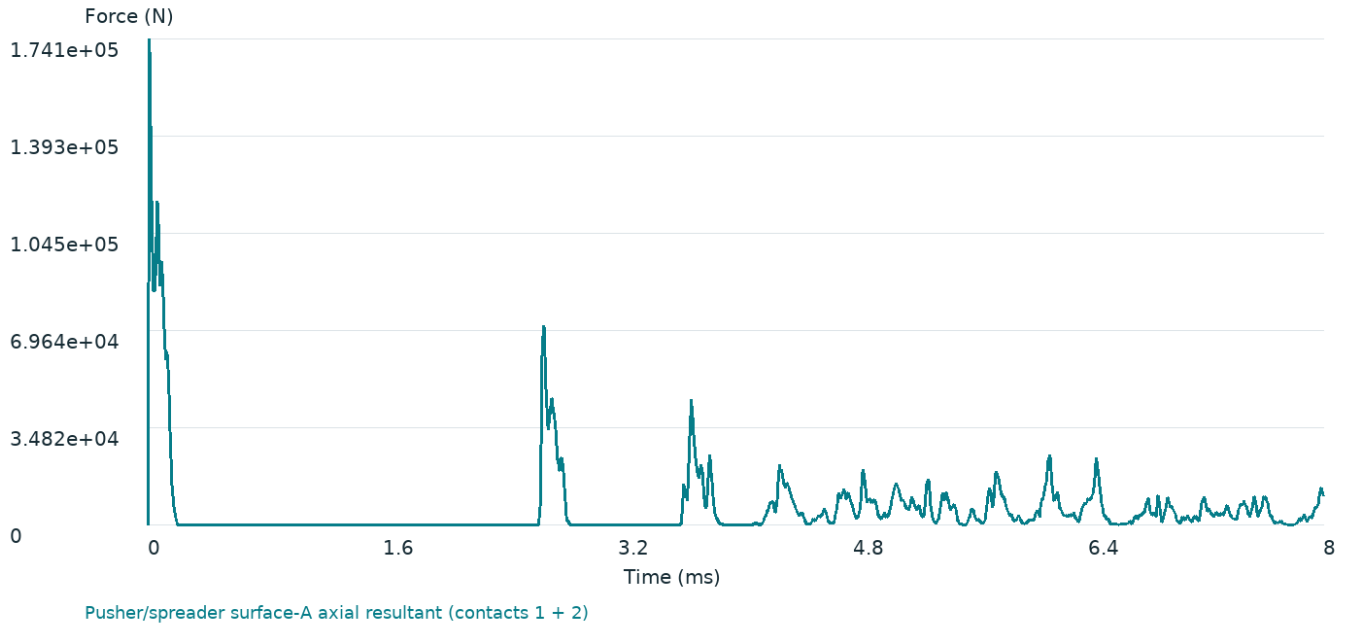
- All axial stopping reaction passes through transient contact; no axial guide constraint.
- Foam rear z fixed, x/y free: frictionless fixed backing; sides unconfined.
- MAT057 synthetic loading and unloading; HU=.05 BETA=0 DAMP=.1 SHAPE=1 FAIL=0. Not a product material model.
- Generic foam anchored at one catalog point; remaining curve chosen for nonbottoming comparative stop.
- Startup-only local mass scaling requested; added mass and inertia error MUST be assessed from solver output.
- Mesh, timestep, energy and momentum require independent assessment.

Solve

Recorded run status	completed_unassessed	
Recorded reached simulation time	7.99998	ms
Recorded requested simulation time	8	ms
Recorded solver wall time	1558.016	s
Recorded termination state	normal_termination_confirmed	

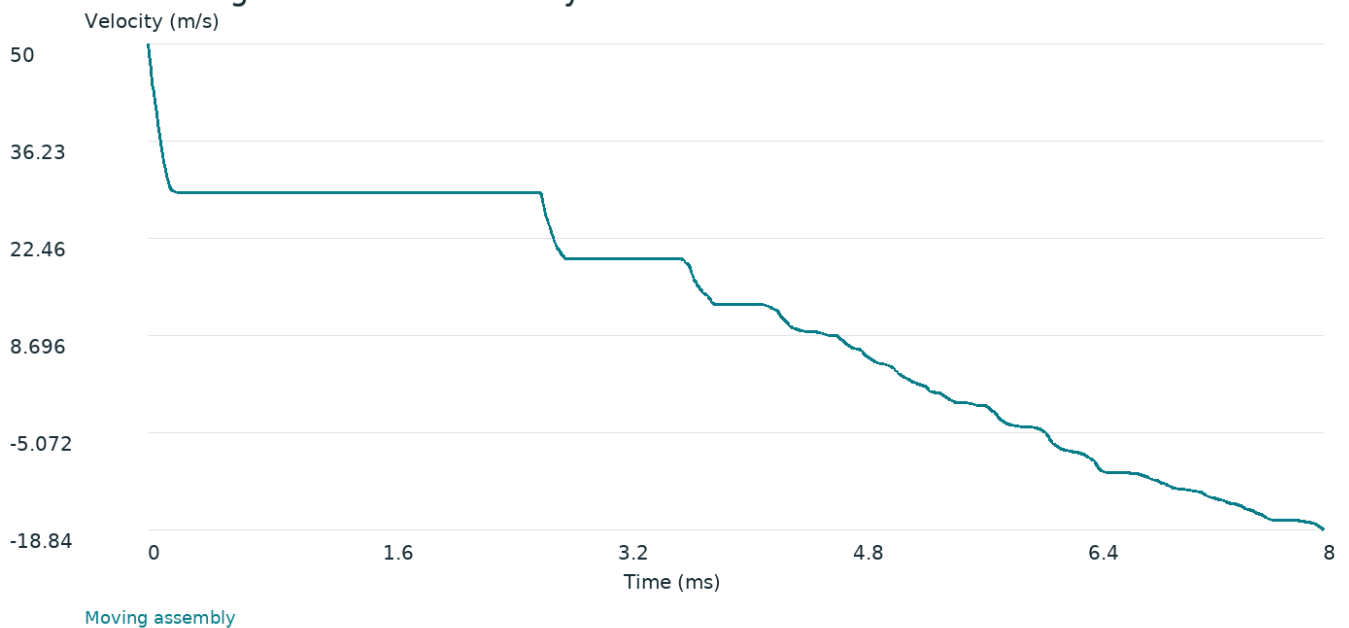
Separate solve charts

Contact force resultant



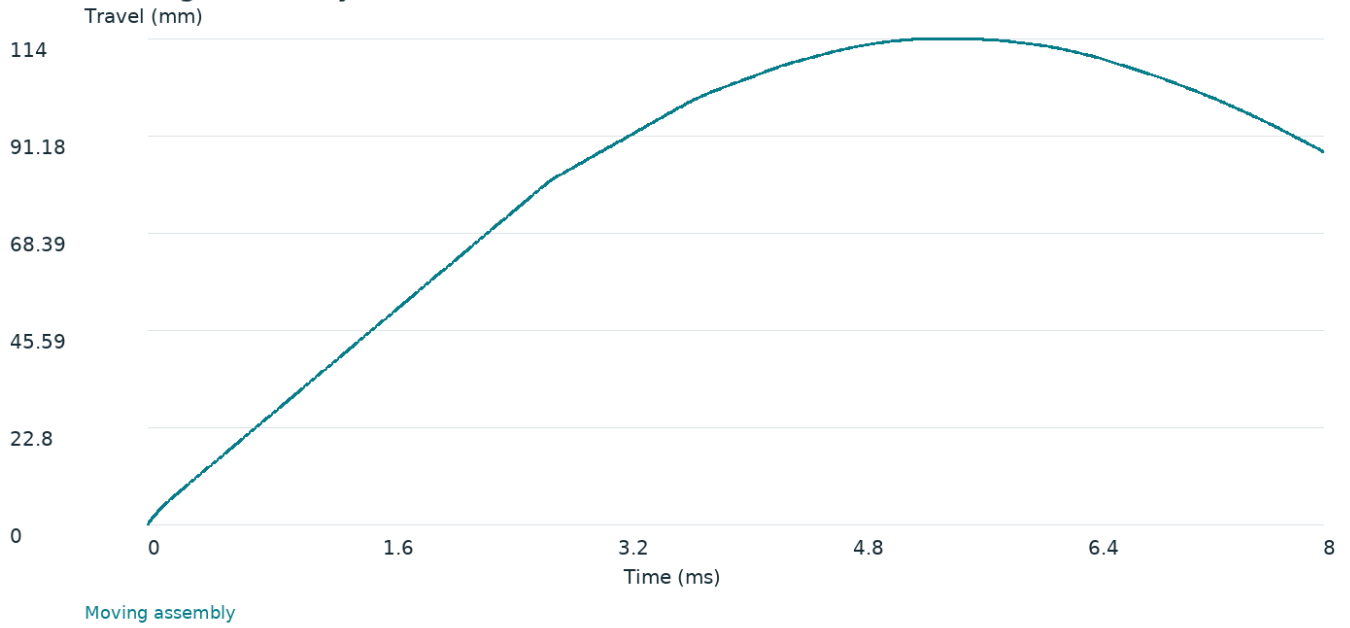
Contact force resultant. projects/compliant_stop_340/cases/baseline-guided-v2/assessment/contact_histories.csv

Mass-weighted forward velocity



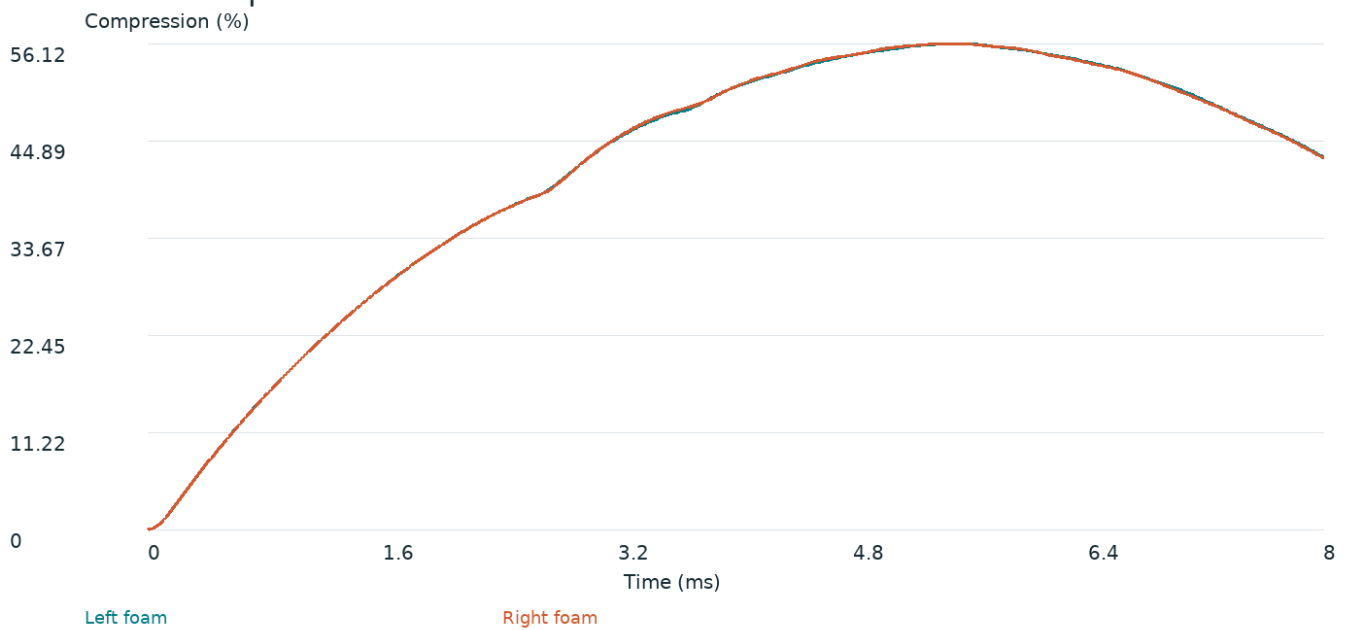
Mass-weighted forward velocity. projects/compliant_stop_340/cases/baseline-guided-v2/assessment/time_histories.csv

Moving assembly travel



Moving assembly travel. projects/compliant_stop_340/cases/baseline-guided-v2/assessment/time_histories.csv

Foam compression



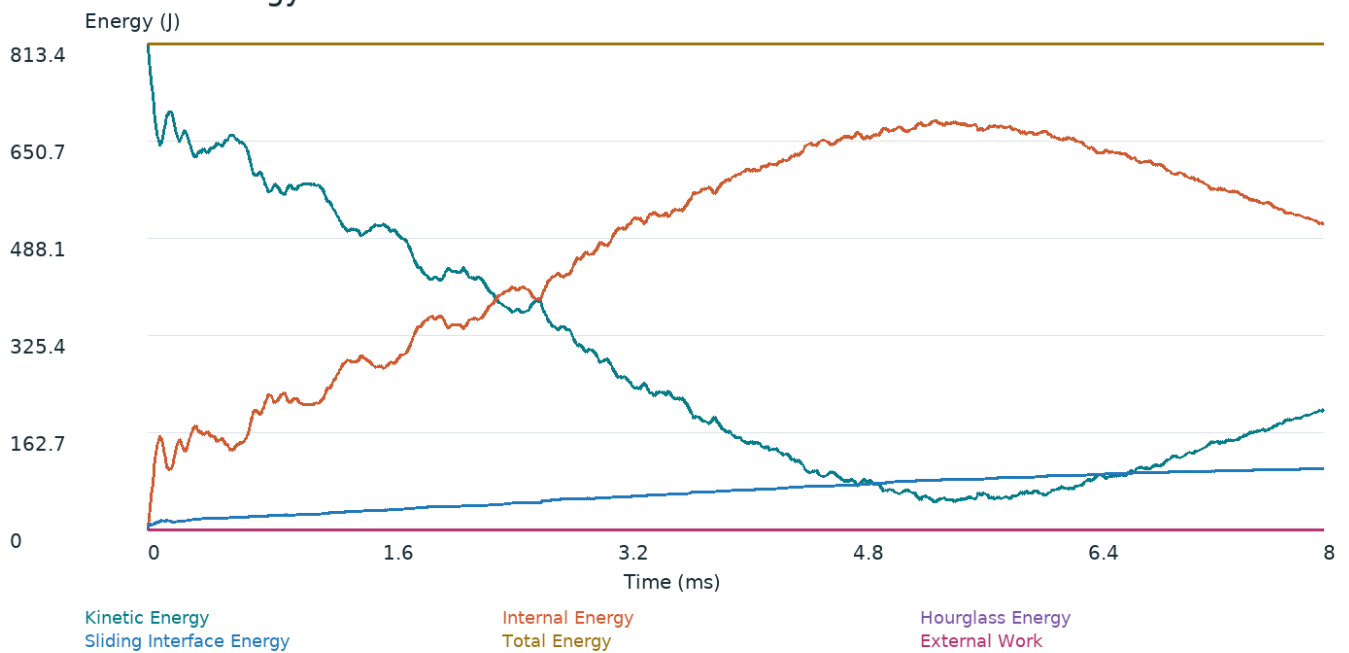
Foam compression. projects/compliant_stop_340/cases/baseline-guided-v2/assessment/time_histories.csv

Regional P99 von Mises stress



Regional P99 von Mises stress. projects/compliant_stop_340/cases/baseline-guided-v2/assessment/stress_histories.csv

Global energy



Global energy. projects/compliant_stop_340/cases/baseline-guided-v2/assessment/global_energy_histories.csv

Results

Run 006 - impact evidence

Evidence status: completed_unassessed. This report describes only extracted evidence from the selected run. Numerical checks alone do not establish physical qualification.

Geometry and model setup

Recorded model item	Value
kind	impact
pusher kg	0.488741
body kg	0.41625
cable cap kg	0.0724904
motors kg	0.1616
moving kg	0.650341
initial ke J	812.926
spreader each kg	0.135871

Solved mesh

Recorded mesh source: mesh_4mm_repaired. Mesh and timestep convergence are not established by these histories.

Motion and stopping event

Signed forward velocity, travel and foam compression are measured histories. Negative forward velocity denotes rebound; a partial history cannot establish the complete stopping event.

Measured result	Value	Unit
Peak stopping force	174104	N
Maximum travel	113.976	mm
Peak regional P99 (cap)	126.725	MPa
Last extracted time	8.00006	ms
Final forward velocity	-18.84	m/s

Regional stress and energy

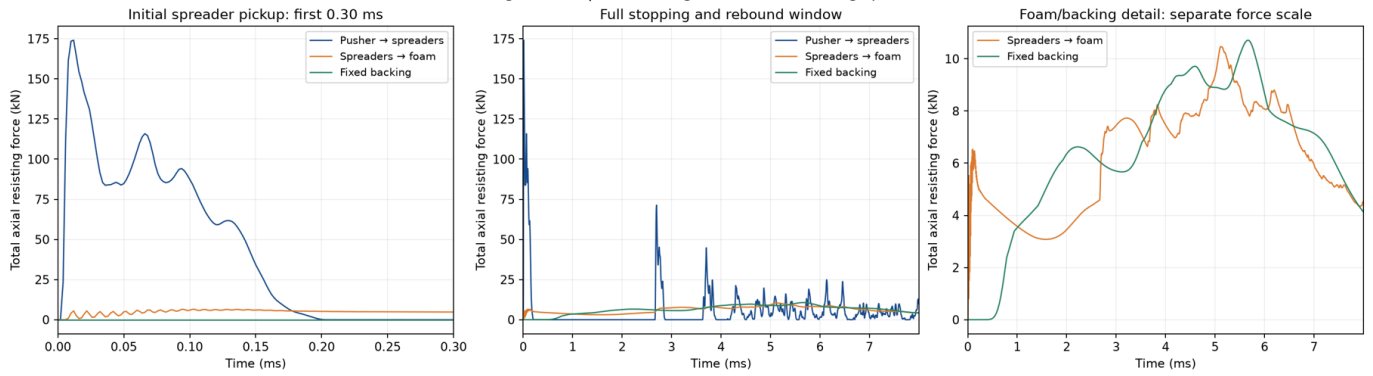
Regional P99 values are elastic demand measures across saved states, not printed-part strength predictions. Force peaks and percentile peaks can occur at different times.

Numerical checks and qualification

Extracted numerical check	Evidence
normal termination	true
requested time reached	true
binary matches nodout	true
moving mass consistent	true
initial energy error below 0p5pct	true
energy balance error below 5pct	true
hourglass below 5pct initial energy	true
sum abs nonfriction contact energy below 5pct initial energy	true
contact impulse momentum error below 5pct	true
no axial spc on moving or guide nodes	true
rigid guides leave only axial translation free	true
no prescribed motion	true
no detected deletion	true
positive sampled foam jacobians	true
first stop captured	true
foam compression below 80pct	true
mass scaling disabled or quantified below 0p1pct moving mass	true
all transverse guide equations satisfied and no axial equation	true

Images, animations and source evidence

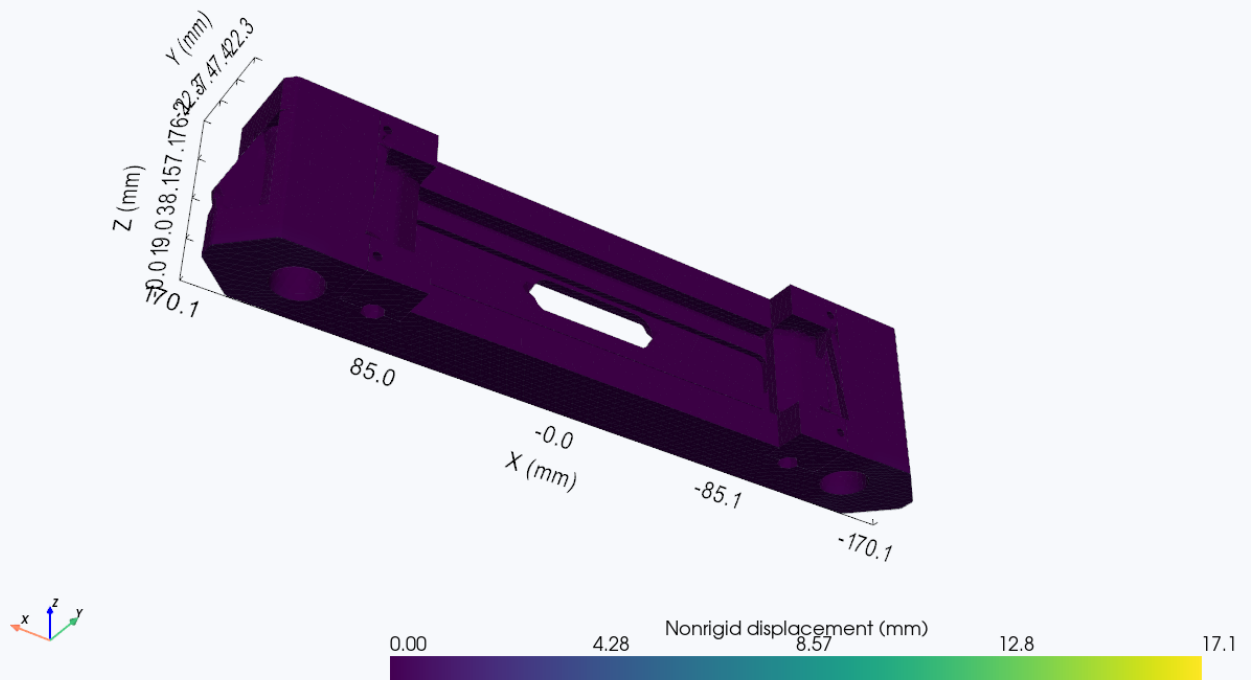
baseline-guided-v2 | Force changes across the moving spreaders



contact-load-path.png

baseline-guided-v2
Rigid motion removed; deformation 1x; original outline gray

t = 0.010 ms

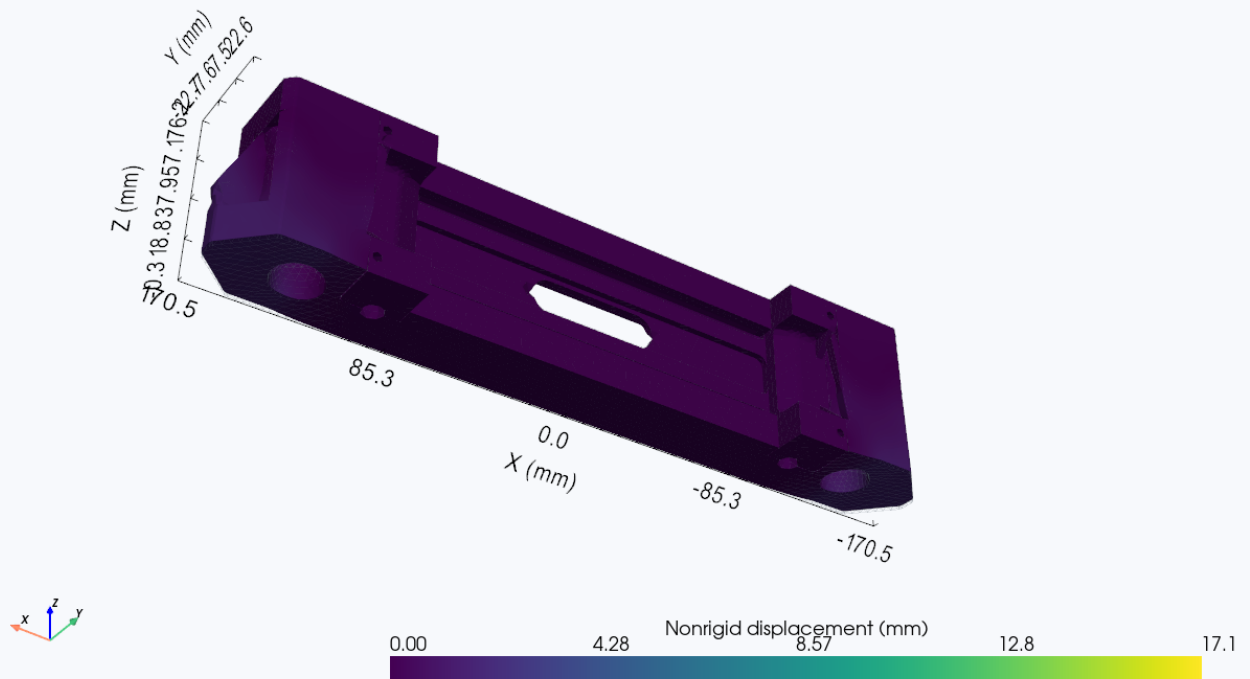


Elastic stress demand / generic foam — no failure prediction

deformation_0001_0.010ms.png

baseline-guided-v2
Rigid motion removed; deformation 1x; original outline gray

t = 0.050 ms

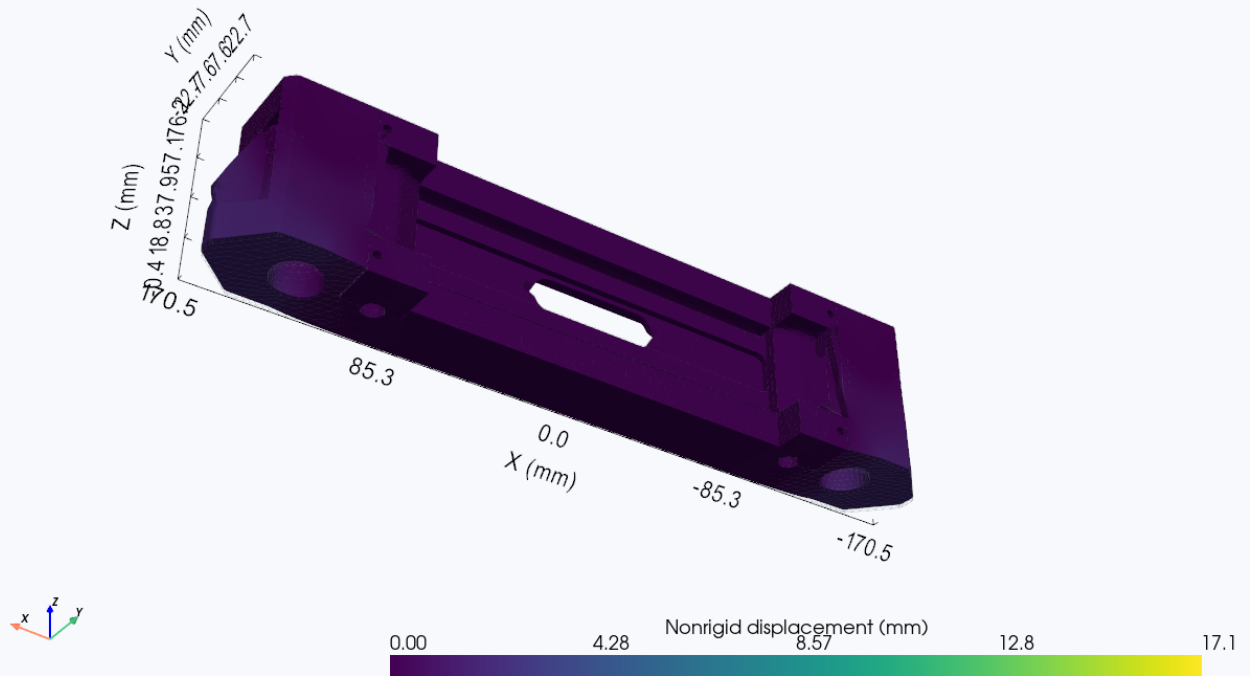


Elastic stress demand / generic foam — no failure prediction

deformation_0005_0.050ms.png

baseline-guided-v2
Rigid motion removed; deformation 1x; original outline gray

t = 0.060 ms

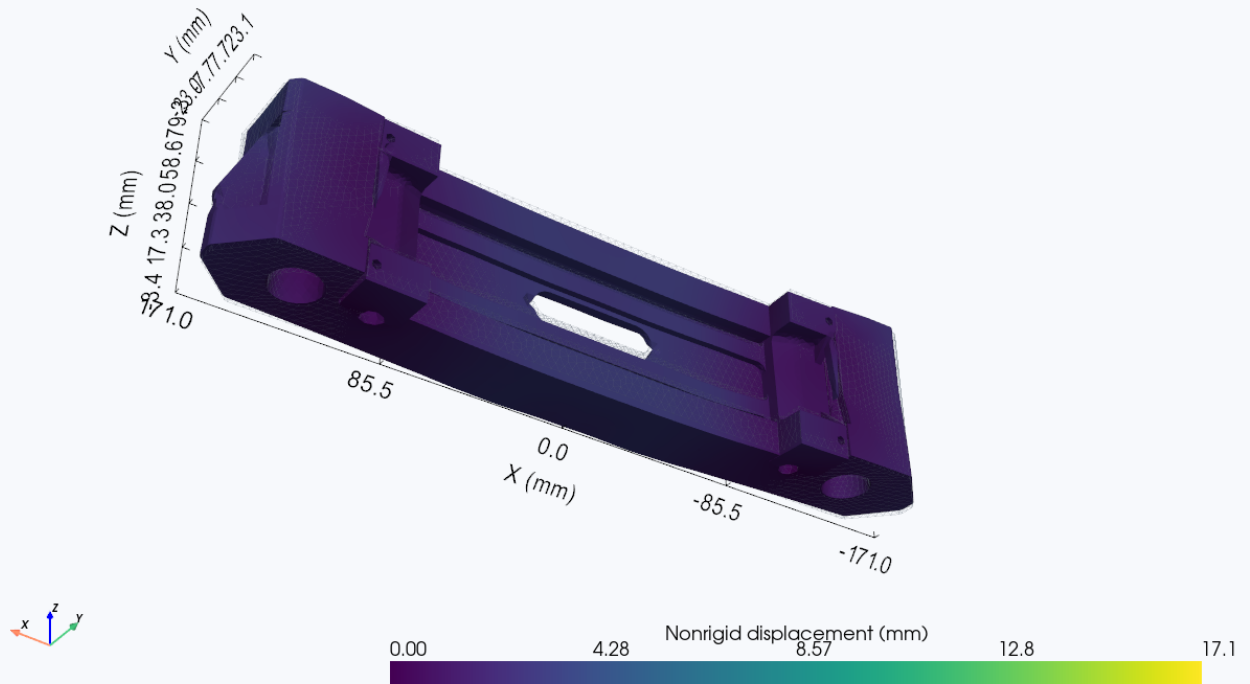


Elastic stress demand / generic foam — no failure prediction

deformation_0006_0.060ms.png

baseline-guided-v2
Rigid motion removed; deformation 1x; original outline gray

t = 0.210 ms

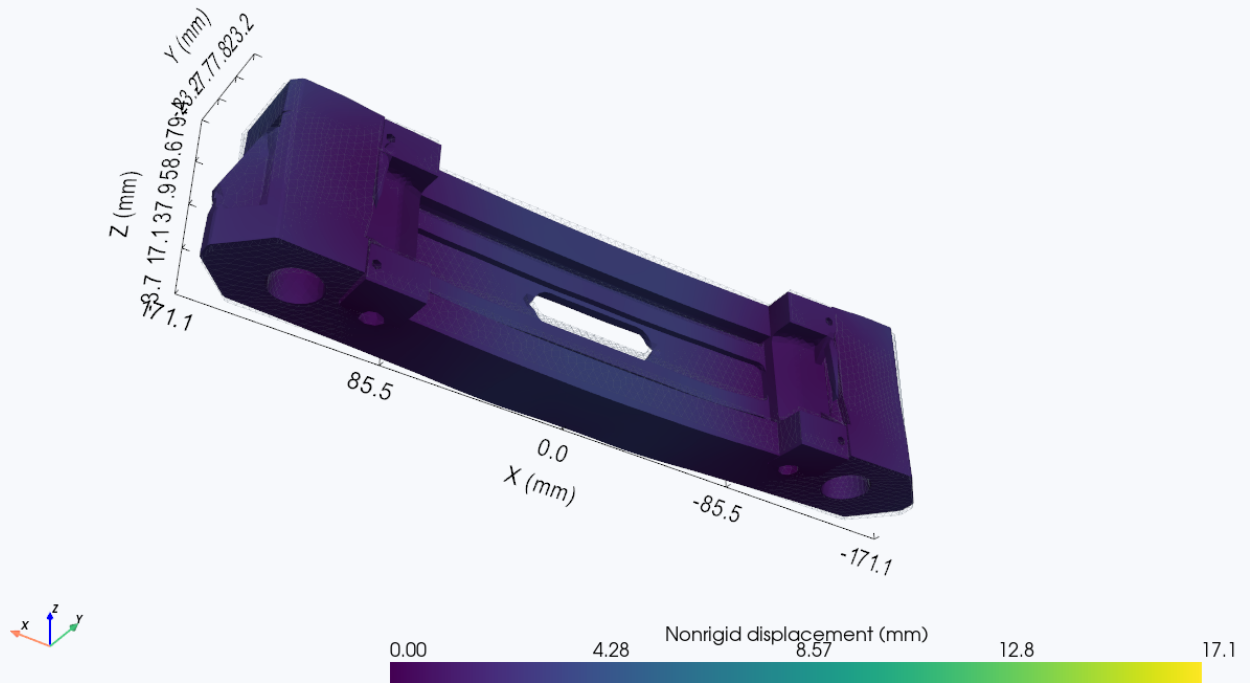


Elastic stress demand / generic foam — no failure prediction

deformation_0021_0.210ms.png

baseline-guided-v2
Rigid motion removed; deformation 1x; original outline gray

t = 0.220 ms

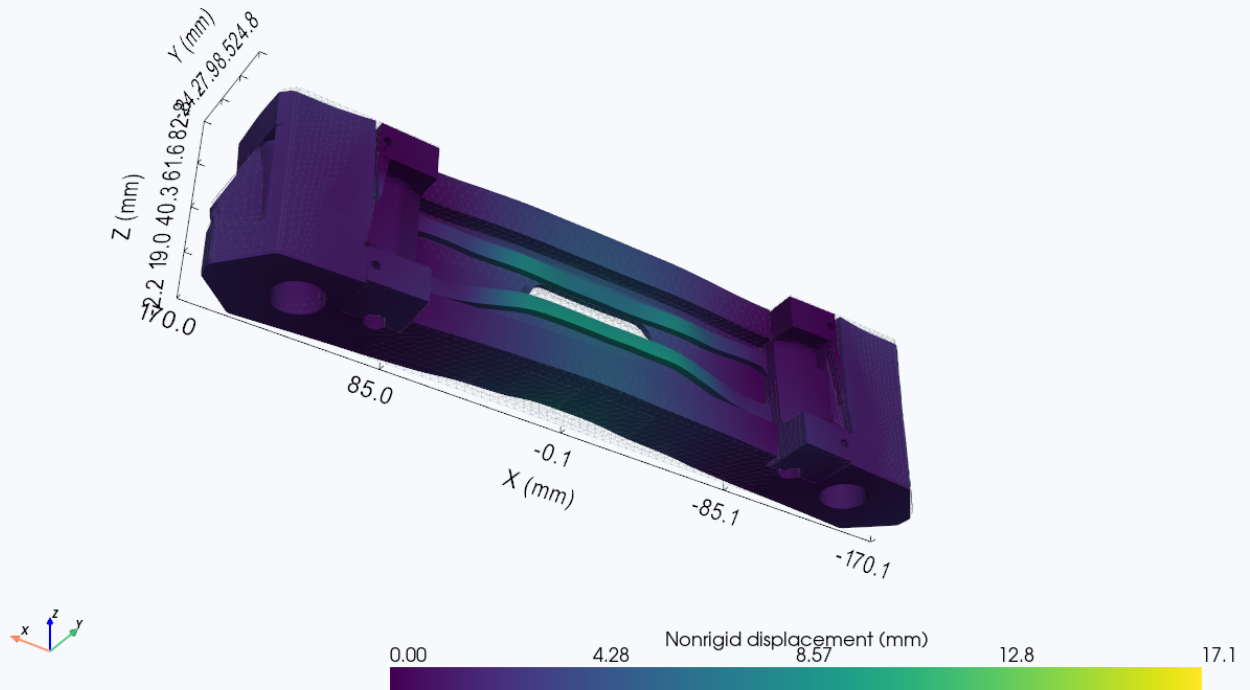


Elastic stress demand / generic foam — no failure prediction

deformation_0022_0.220ms.png

baseline-guided-v2
Rigid motion removed; deformation 1x; original outline gray

t = 0.820 ms

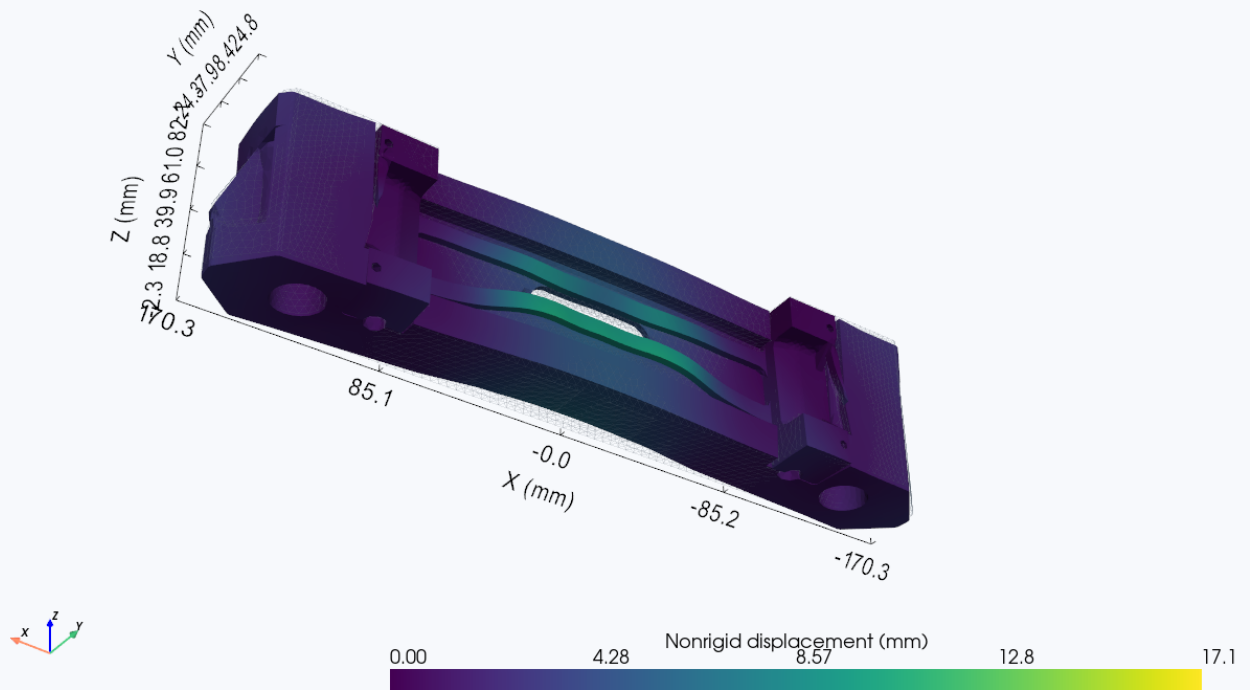


Elastic stress demand / generic foam — no failure prediction

deformation_0082_0.820ms.png

baseline-guided-v2
Rigid motion removed; deformation 1x; original outline gray

t = 0.850 ms

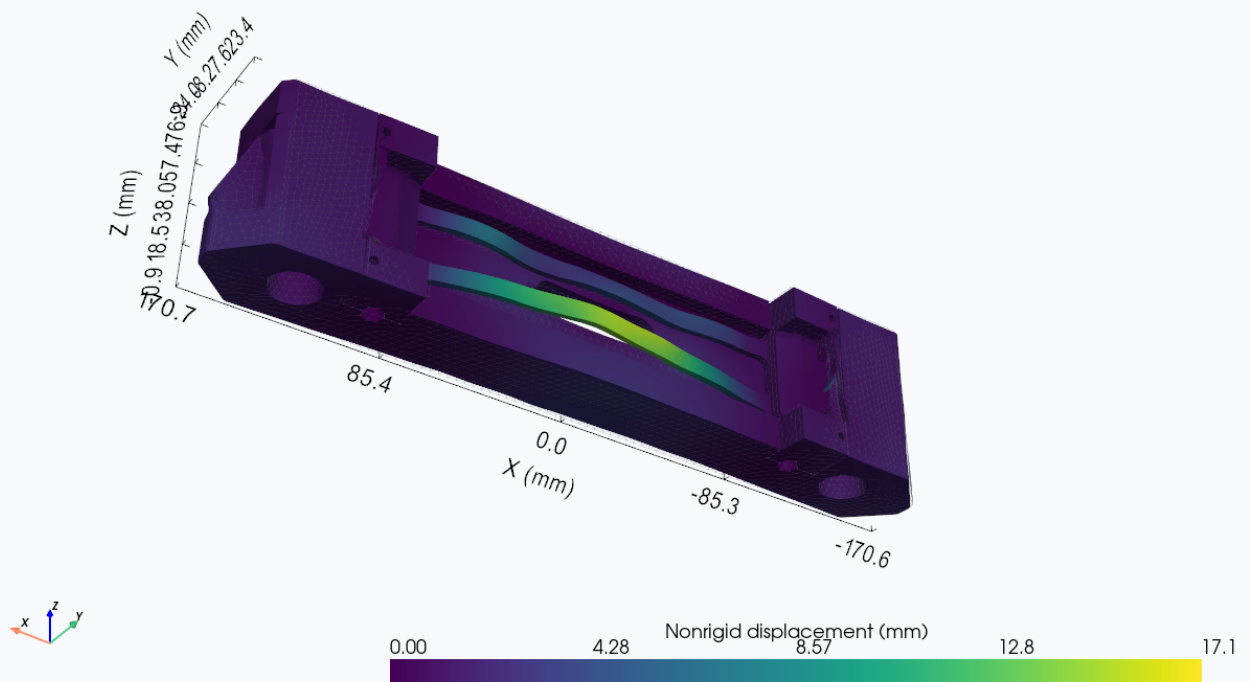


Elastic stress demand / generic foam — no failure prediction

deformation_0085_0.850ms.png

baseline-guided-v2
Rigid motion removed; deformation 1x; original outline gray

t = 5.430 ms

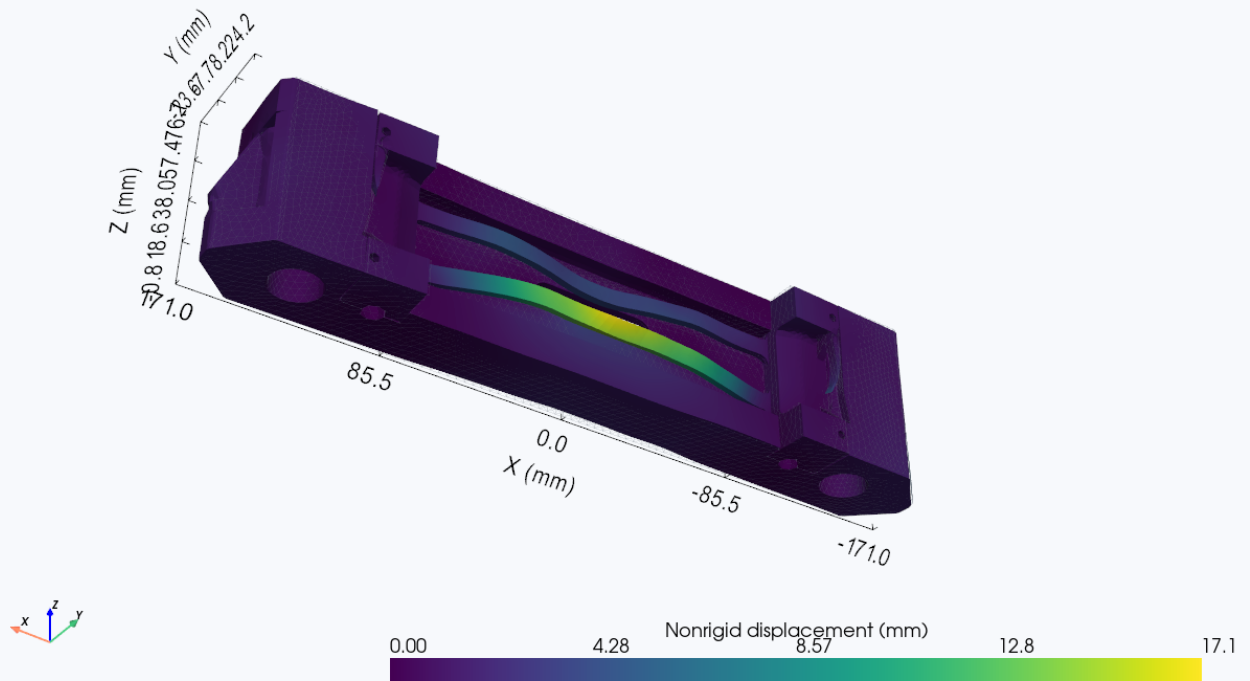


Elastic stress demand / generic foam — no failure prediction

deformation_0543_5.430ms.png

baseline-guided-v2
Rigid motion removed; deformation 1x; original outline gray

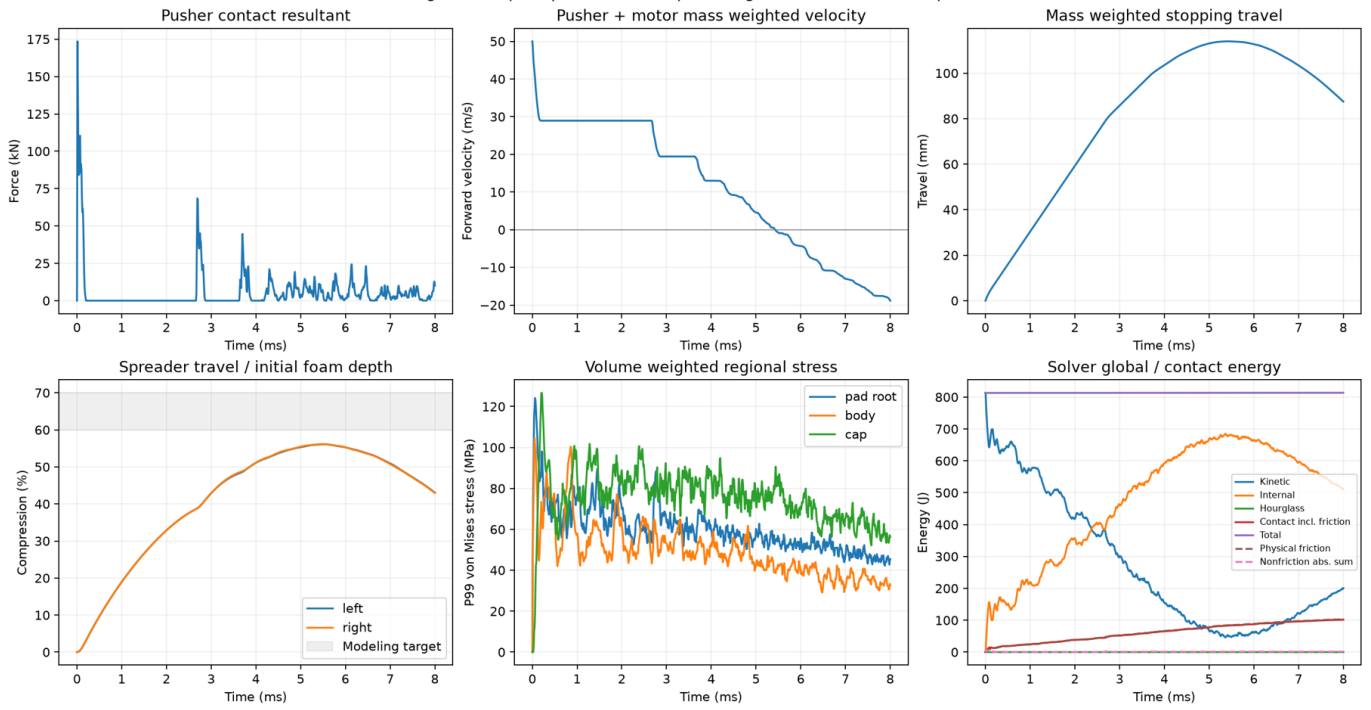
t = 5.540 ms



Elastic stress demand / generic foam — no failure prediction

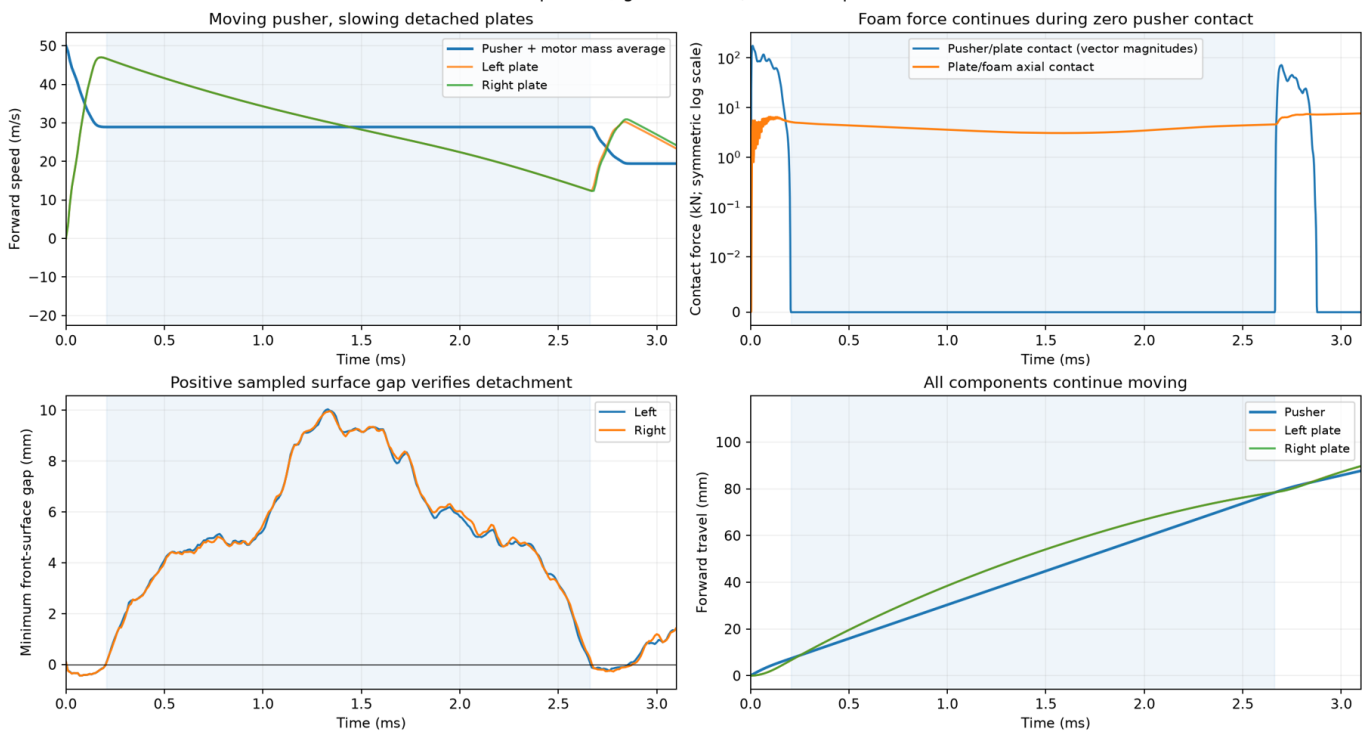
deformation_0554_5.540ms.png

baseline-guided-v2 | comparative elastic pusher / generic foam; no failure prediction



histories.png

baseline-guided-v2: measured coasting interval 0.208–2.662 ms (shaded)
Elastic comparison / generic foam; no failure prediction

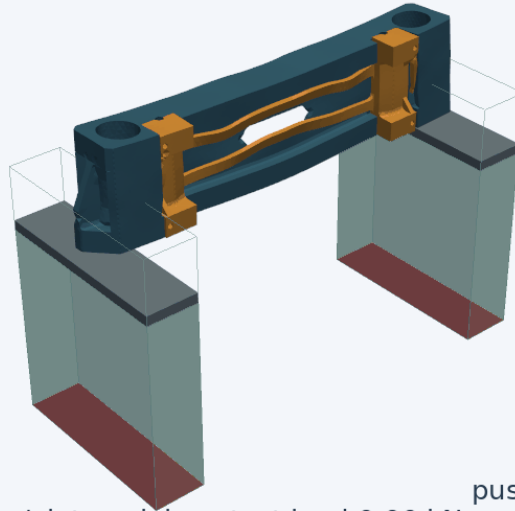


plateau_diagnostic.png

WORLD MOTION
Time 1.380 ms; impact along -Z

baseline-guided-v2

- ▲ Generic foam
- ▲ Fixed backing
- ▲ Aluminum plates
- ▲ Pusher body
- ▲ Cable cap



COASTING

Measured pusher/plate axial contact load 0.00 kN

Pusher forward speed 28.93 m/s

travel 41.4 mm

Foam compression L/R 24.9% / 24.9%

True displacement 1x

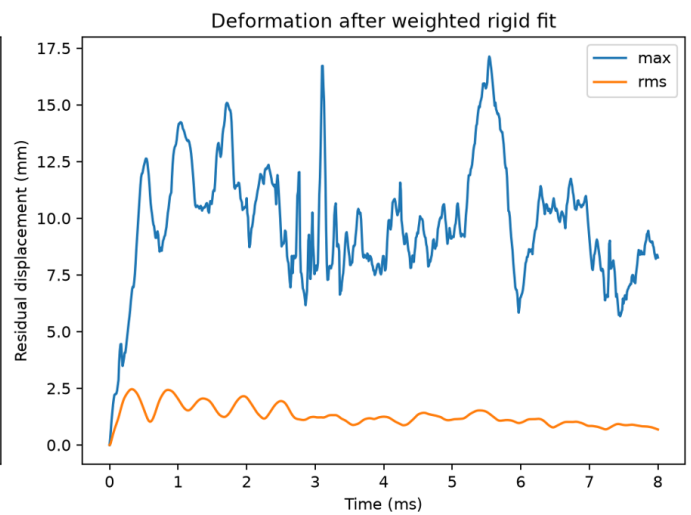
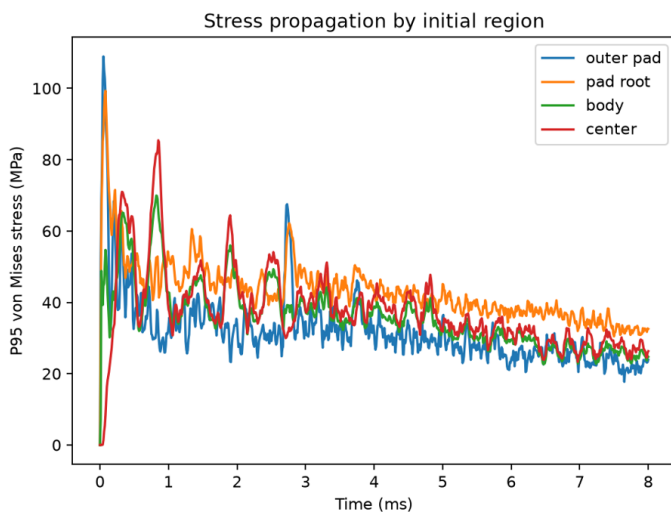
Foam initial outline in gray; axes and geometry in mm

pusher separated from plates



component colors, not a stress contour

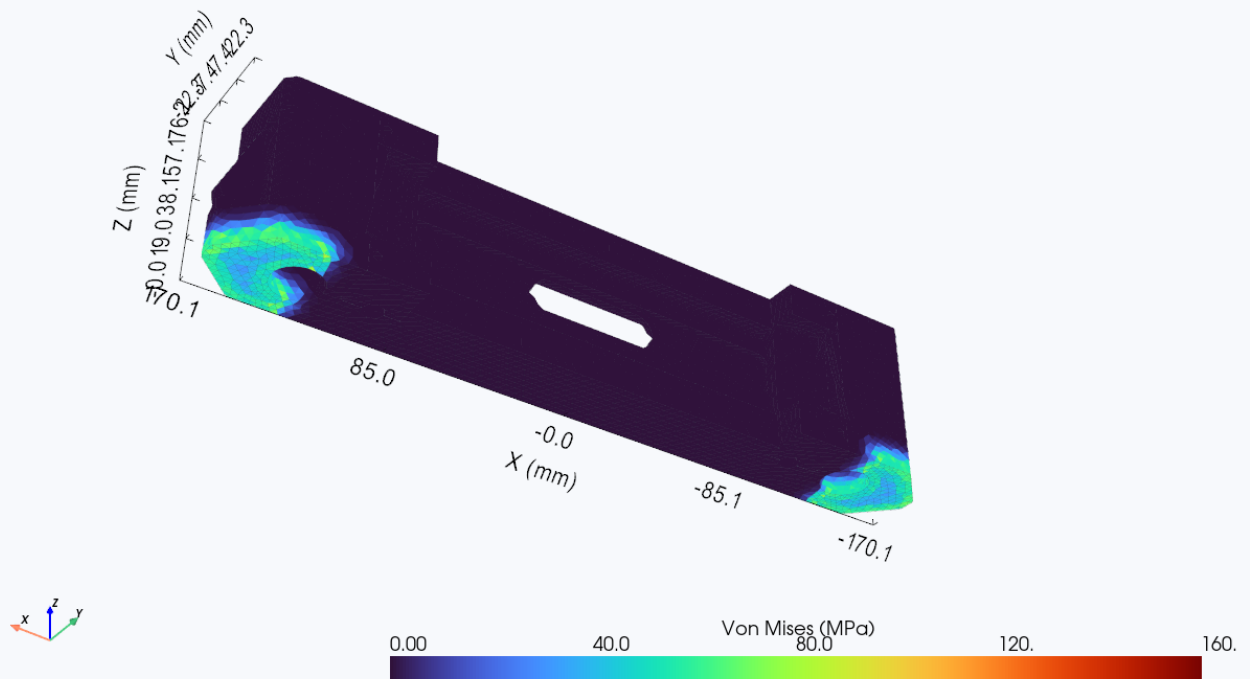
plateau_diagnostic_world_frame.png



stress-propagation-deformation.png

baseline-guided-v2
Rigid motion removed; deformation 1x; original outline gray

t = 0.010 ms

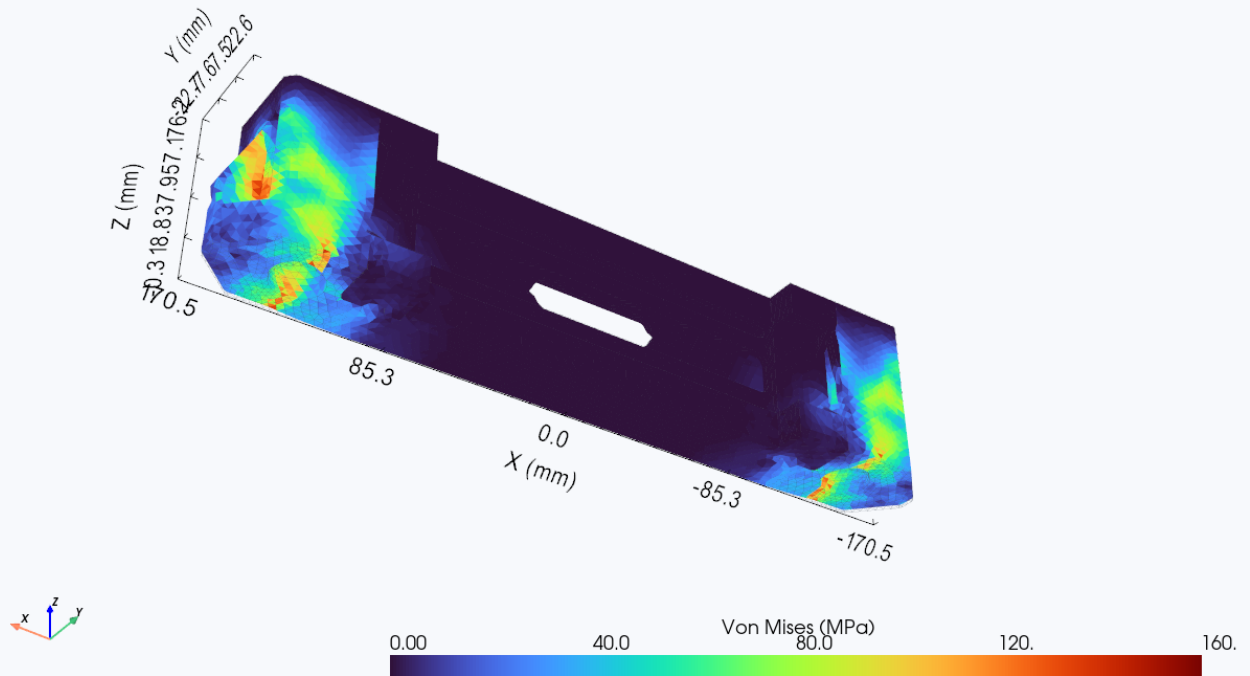


Display saturates at 160 MPa; raw peak 666.5 MPa elastic proxy

stress_0001_0.010ms.png

baseline-guided-v2
Rigid motion removed; deformation 1x; original outline gray

t = 0.050 ms

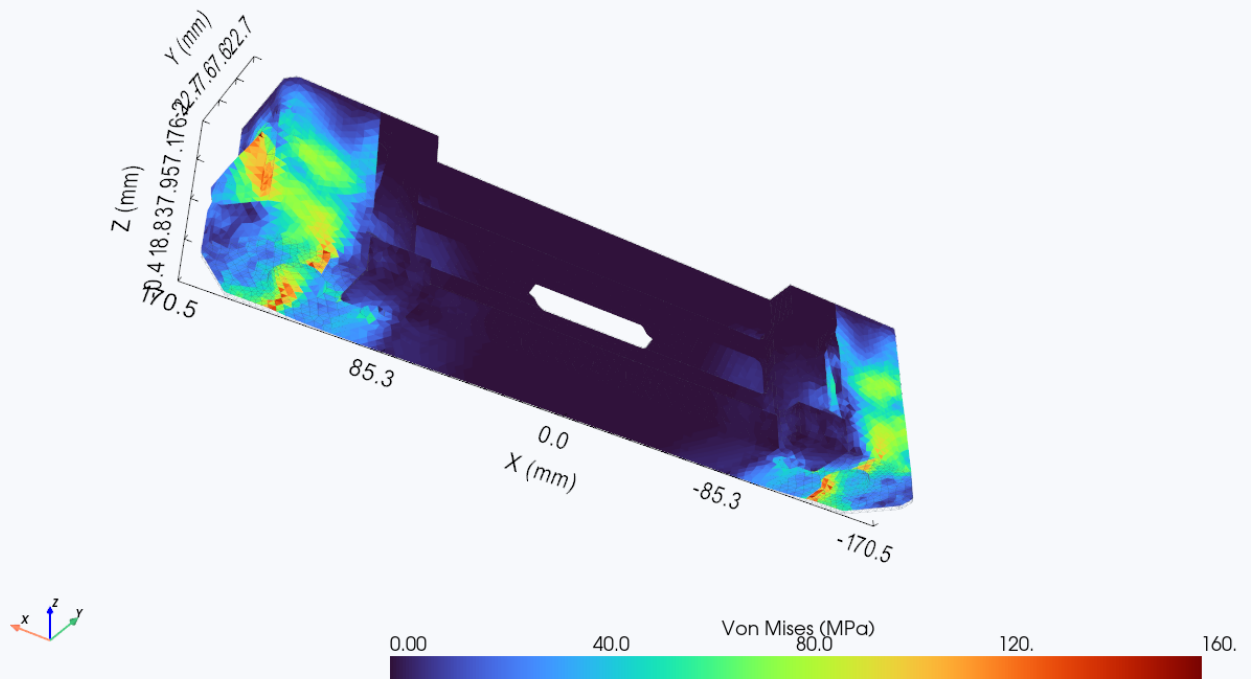


Display saturates at 160 MPa; raw peak 666.5 MPa elastic proxy

stress_0005_0.050ms.png

baseline-guided-v2
Rigid motion removed; deformation 1x; original outline gray

t = 0.060 ms

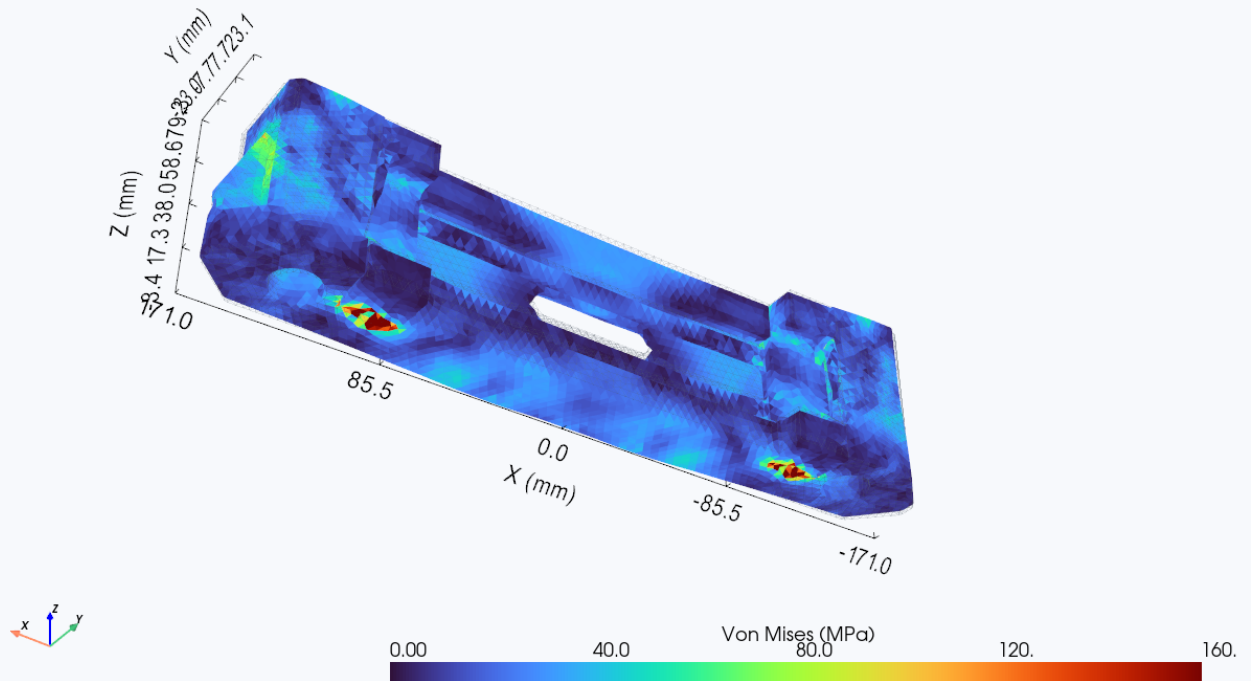


Display saturates at 160 MPa; raw peak 666.5 MPa elastic proxy

stress_0006_0.060ms.png

baseline-guided-v2
Rigid motion removed; deformation 1x; original outline gray

t = 0.210 ms

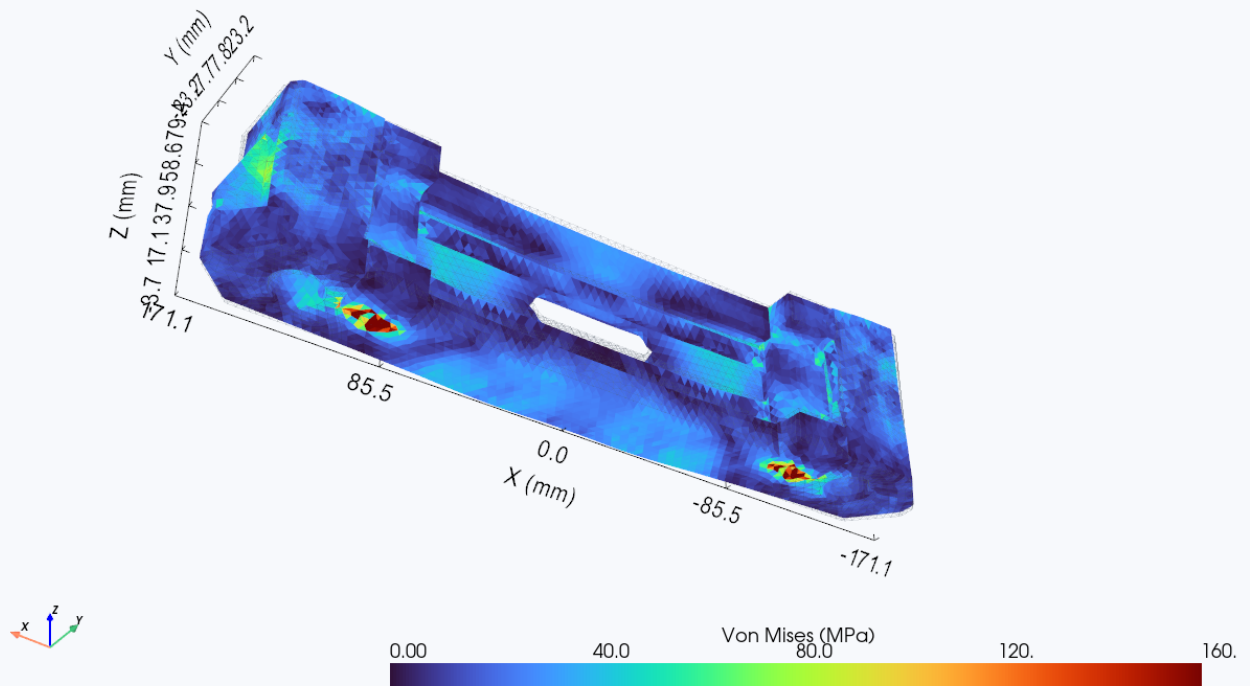


Display saturates at 160 MPa; raw peak 666.5 MPa elastic proxy

stress_0021_0.210ms.png

baseline-guided-v2
Rigid motion removed; deformation 1x; original outline gray

t = 0.220 ms

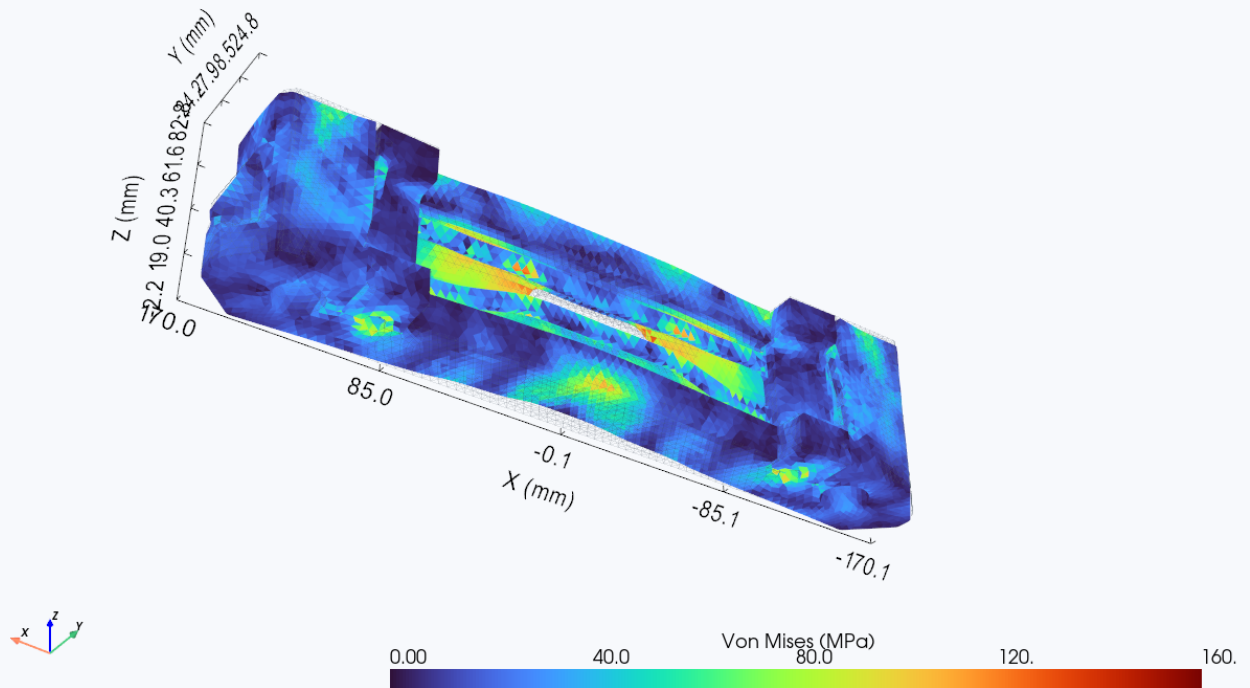


Display saturates at 160 MPa; raw peak 666.5 MPa elastic proxy

stress_0022_0.220ms.png

baseline-guided-v2
Rigid motion removed; deformation 1x; original outline gray

t = 0.820 ms

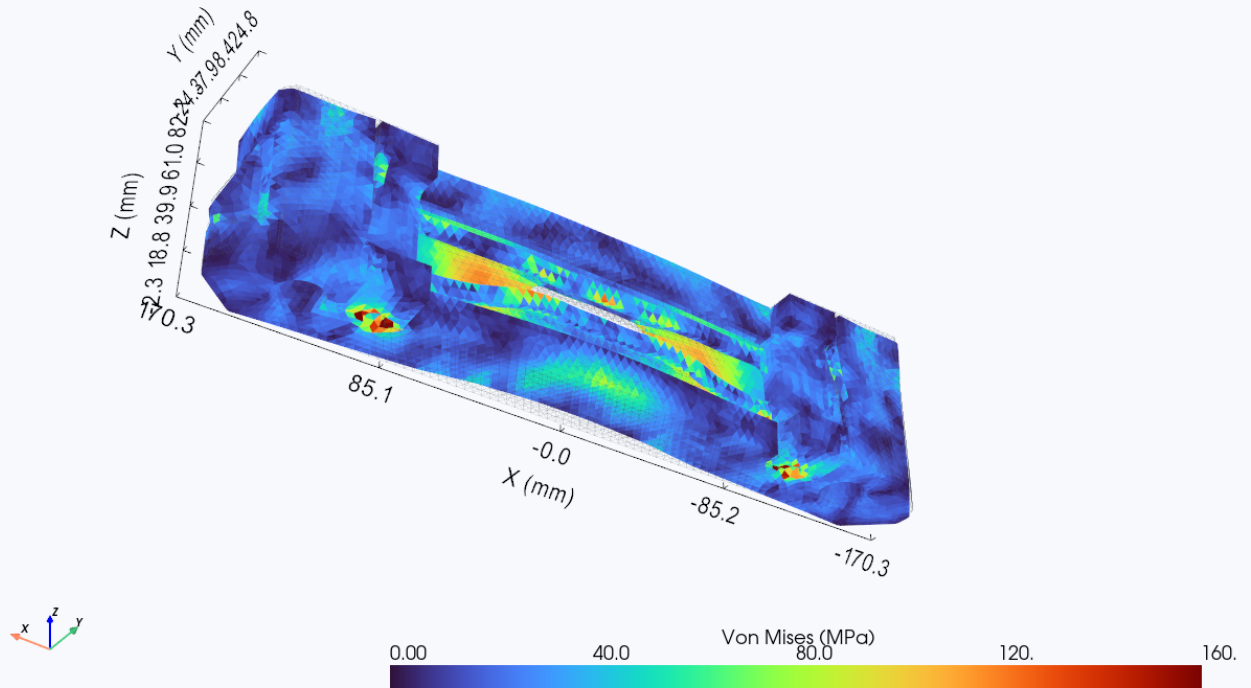


Display saturates at 160 MPa; raw peak 666.5 MPa elastic proxy

stress_0082_0.820ms.png

baseline-guided-v2
Rigid motion removed; deformation 1x; original outline gray

t = 0.850 ms

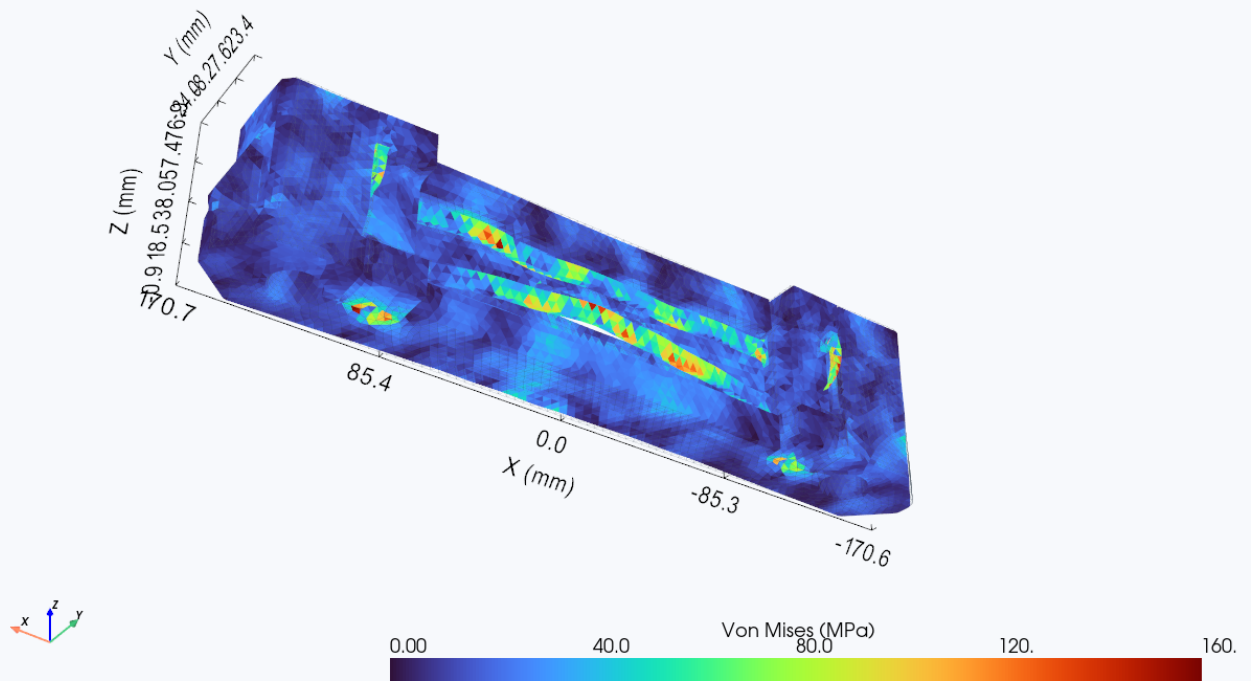


Display saturates at 160 MPa; raw peak 666.5 MPa elastic proxy

stress_0085_0.850ms.png

baseline-guided-v2
Rigid motion removed; deformation 1x; original outline gray

t = 5.430 ms

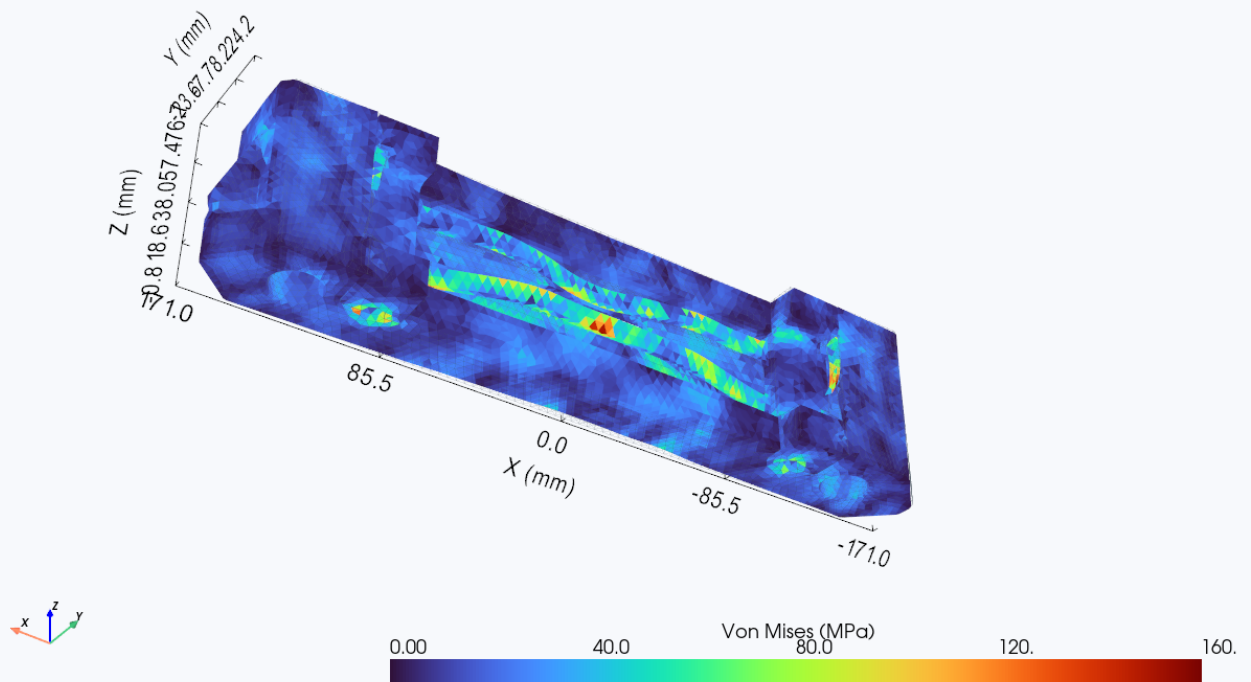


Display saturates at 160 MPa; raw peak 666.5 MPa elastic proxy

stress_0543_5.430ms.png

baseline-guided-v2
Rigid motion removed; deformation 1x; original outline gray

t = 5.540 ms



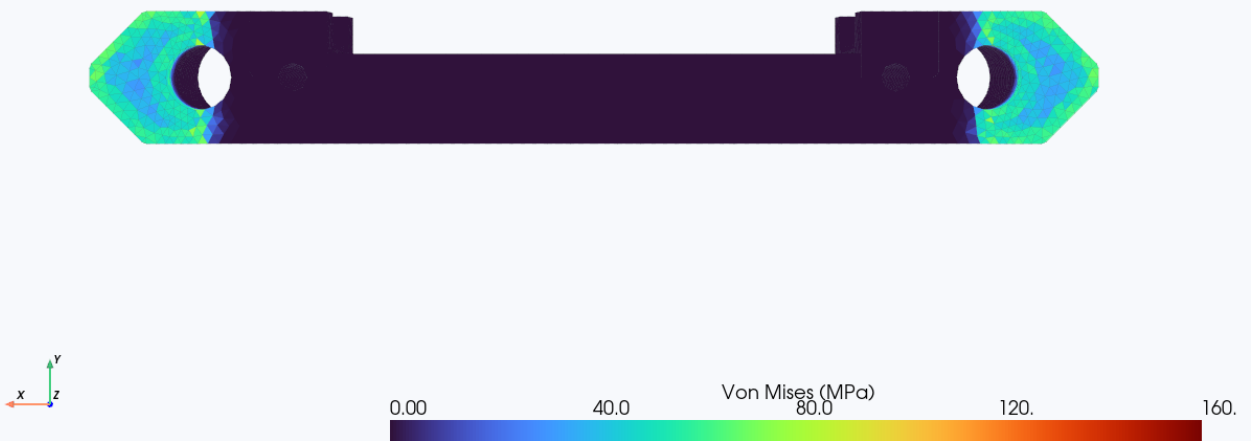
Display saturates at 160 MPa; raw peak 666.5 MPa elastic proxy

stress_0554_5.540ms.png

baseline-guided-v2
Rigid motion removed; deformation 1x; original outline gray

t = 0.010 ms

IMPACT FACE (-Z)



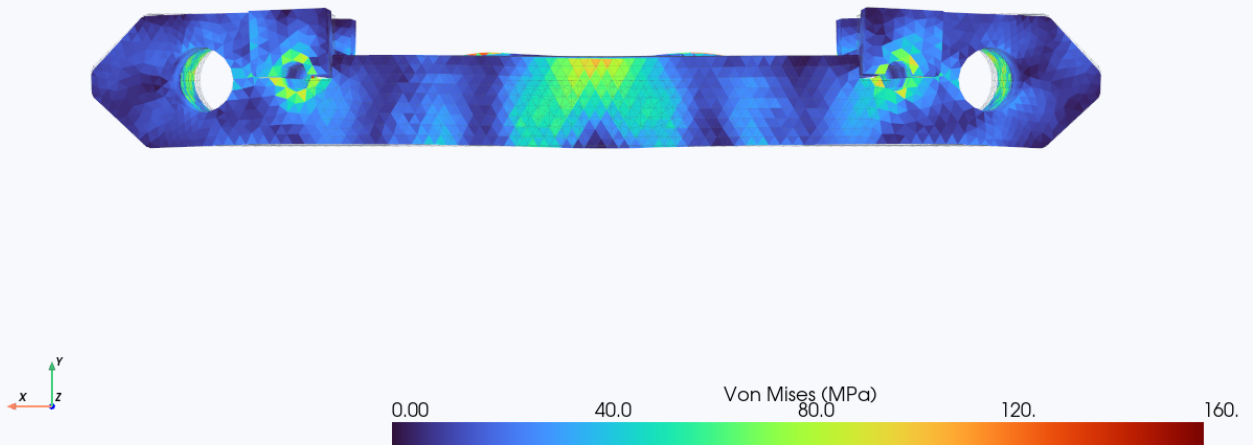
Display saturates at 160 MPa; raw peak 666.5 MPa elastic proxy

stress_impact_0001_0.010ms.png

baseline-guided-v2
Rigid motion removed; deformation 1x; original outline gray

t = 0.820 ms

IMPACT FACE (-Z)



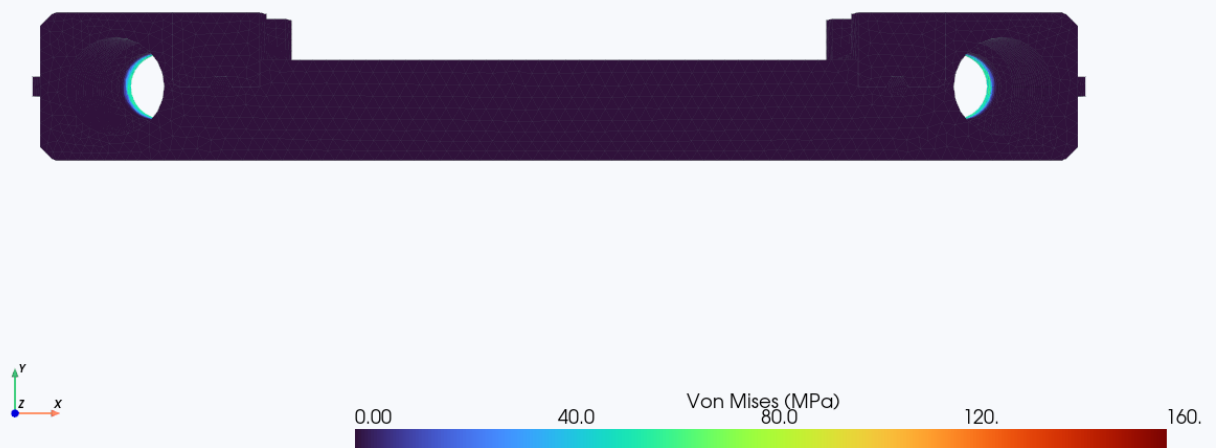
Display saturates at 160 MPa; raw peak 666.5 MPa elastic proxy

stress_impact_0082_0.820ms.png

baseline-guided-v2
Rigid motion removed; deformation 1x; original outline gray

t = 0.010 ms

TOP (+Z)



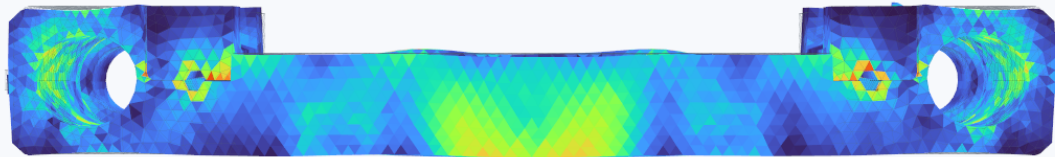
Display saturates at 160 MPa; raw peak 666.5 MPa elastic proxy

stress_top_0001_0.010ms.png

baseline-guided-v2
Rigid motion removed; deformation 1x; original outline gray

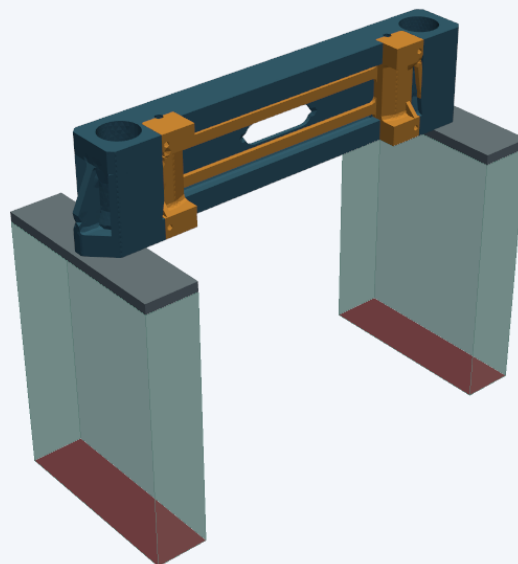
t = 0.820 ms

TOP (+Z)



Display saturates at 160 MPa; raw peak 666.5 MPa elastic proxy

stress_top_0082_0.820ms.png



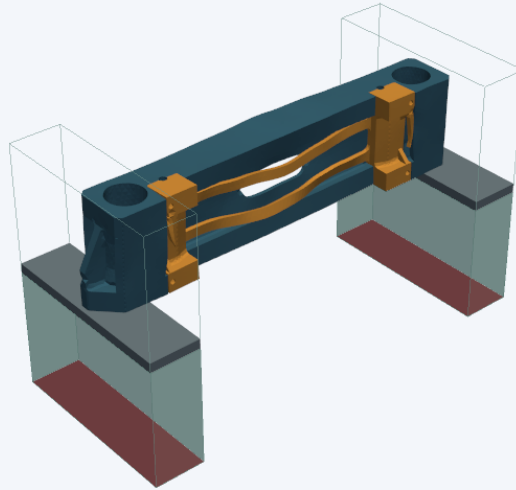
- ▲ Generic foam
- ▲ Fixed backing
- ▲ Aluminum plates
- ▲ Pusher body
- ▲ Cable cap

Speed 50.00 m/s
Compression L/R
-0.0% / -0.0%
Travel -0.0 mm
Load 0.00 kN
Time 0.000 ms

world_0000_0.000ms.png

- ▲ Generic foam
- ▲ Fixed backing
- ▲ Aluminum plates
- ▲ Pusher body
- ▲ Cable cap

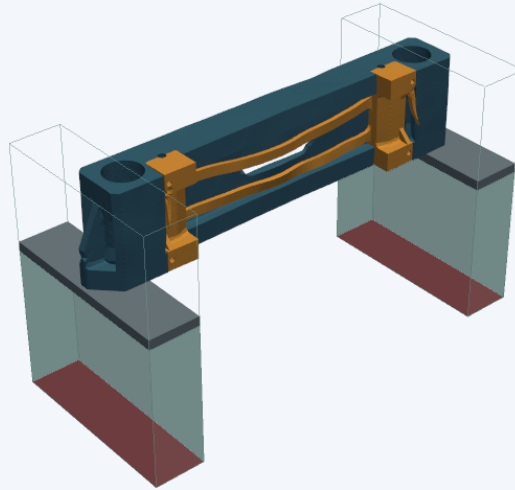
Speed 12.98 m/s
Compression L/R
51.9% / 52.0%
Travel 105.3 mm
Load 0.52 kN
Time 4.140 ms



world_0414_4.140ms.png

- ▲ Generic foam
- ▲ Fixed backing
- ▲ Aluminum plates
- ▲ Pusher body
- ▲ Cable cap

Speed -18.84 m/s
 Compression L/R
 43.0% / 42.9%
 Travel 87.5 mm
 Load 10.11 kN
 Time 8.000 ms



world_0801_8.000ms.png

- cable_guide_motion.csv
- contact_energy_by_interface.csv
- contact_energy_components.csv
- contact_histories.csv
- final_render_audit.json
- global_energy_histories.csv
- load_path_peaks.json
- maximum_deformation_direction.json
- metrics.json
- metrics_before_final_render.json
- motion_render_audit.json
- part_energy_histories.csv
- plateau_diagnostic.json

- `sampled_planar_penetration.csv`
- `spc_histories.csv`
- `stress_histories.csv`
- `stress_restyle_audit.json`
- `time_histories.csv`
- `weighted_guide_constraints.csv`

Generated from selected-run extracted evidence; no authored report is available. Animation figures and their captions are excluded from this PDF; animations remain available in the standalone HTML.